What explains movement across the AI value chain?

A factor-model study of the stocks, data and market signals behind AI infrastructure.

Supervisor: Ottucsák György
Pete Zétény, Sánta Dávid, Halász Erik

From broad market movements to the power and compute cycle

A stepwise linear model separates market drivers

We want broad index returns to act as factors; each step asks what a new group explains beyond the previous one.

A SIMPLE REGRESSION

Stock return = market exposure + factor exposures + unexplained movement

1. **MARKET** — Broad index
2. **+ MACRO / STYLE** — Rates, dollar, oil
3. **+ AI-ADJACENT** — Power, grid, uranium
4. **+ AI-CLOSE** — Semis, software, robotics

STEPWISE LINEAR REGRESSION

Fit a market-only model, then add factor blocks in sequence. Compare out-of-sample R-squared at each step to see whether the added information improves fit.

ORTHOGONAL FACTORS

Later factors are projected on earlier ones; the overlapping part is removed. Fit this projection on estimation data, then apply it unchanged to the holdout. The order matters.

ETF prices are practical proxies for index / sector returns when a direct index series is not convenient to download.

Daily stock, ETF and macro data build the test universe

Downloaded price history runs from 4 Jan 2016 through 1 Oct 2026; the AI chain spans seven layers.

TICKERS WITH DATA

621 / 623

2 requested tickers were empty

DATA WINDOW

2016-2026

Daily; through 1 Oct 2026

AI-CHAIN NAMES

39 stocks

Plus 8 non-AI controls

DOWNLOADS

526 | 77 | 16

US equities | ETFs | macro series

ALL 39 AI-CHAIN STOCKS IN THE BASKET

CHIPS

NVDA AMD AVGO TSM MRVL MU ARM

NETWORK / SERVERS

ANET CRDO COHR SMCI DELL CLS

HYPERSCALERS / SOFTWARE

MSFT GOOGL AMZN META ORCL

DATA CENTERS

EQIX DLR CRWV NBIS IREN APLD

POWER GENERATION

VST CEG NRG TLN

GRID / ELECTRICAL

GEV ETN VRT PWR HUBB POWL

NUCLEAR / URANIUM

CCJ OKLO SMR BWXT LEU

PLUS CONTROLS (SLIDE 6)

KO PG WMT MCD JNJ JPM XOM CAT

Ticker categories are separated from the model's target basket.

MARKET DATA vs. MACRO DATA

MARKET / SECTOR

Index & ETF prices; e.g., SPY, QQQ, IWM, SMH

MACRO SERIES

16 series, including Treasury yields and rates

ETF = exchange-traded fund: a fund traded on an exchange that holds a basket of assets.

ETF / index return correlations in data QA



High correlation supports proxies; it does not make an ETF an index.

Adding factor groups improves fit across most layers

The 12-factor model adds information in stages; the biggest gains differ by part of the AI chain.

	Market only	+ Macro / style	+ AI-adjacent	+ AI-close
Chips	0.57	0.70	0.76	0.91
Networking / servers	0.45	0.55	0.65	0.71
Hyperscalers / software	0.61	0.65	0.67	0.74
Data centers	0.27	0.31	0.40	0.40
Power generation	0.26	0.30	0.83	0.84
Grid / electrical	0.41	0.47	0.76	0.77
Nuclear / uranium	0.24	0.27	0.68	0.68
Core basket	0.64	0.74	0.89	0.92

Factor blocks: MKT; then D10Y, USD, OIL, LOWVOL; then POWER, GRID, URANIUM; then SEMI, SOFT, ROBO, MOM.

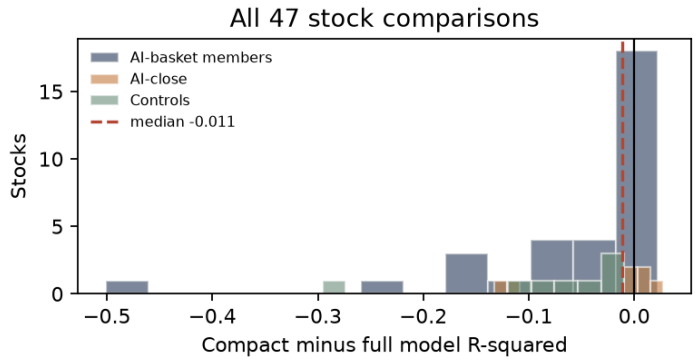
Rolling 252-trading-day OLS; estimate 2017-2023, holdout 2024-2026. R-squared is holdout fit vs. a trailing-mean benchmark. Core basket: 27 AI-chain stocks with >=98% estimation-window coverage, equal-weighted and daily rebalanced.

A five-factor version keeps the model interpretable

Current compact set: MKT, D10Y, LOWVOL, POWER and SEMI; the stock-level comparison figure is shown here.

HOLDOUT R-SQUARED	M0 MKT	M1 +D10Y	M2 +LOWVOL	M3 +POWER	M4 +SEMI
Chips	0.57	0.57	0.69	0.71	0.91
Network / servers	0.45	0.45	0.55	0.61	0.70
Hyperscalers	0.61	0.61	0.66	0.65	0.70
Data centers	0.26	0.26	0.31	0.35	0.36
Power generation	0.26	0.26	0.30	0.82	0.82
Grid / electrical	0.41	0.41	0.48	0.61	0.66
Nuclear / uranium	0.24	0.24	0.27	0.36	0.37
Core basket	0.64	0.64	0.74	0.83	0.89

COMPACT - FULL R-SQUARED: 47 STOCKS



HIGHER COMPACT FIT

15 / 47

Median gap: -0.011

WITHIN 0.05 R2

31 / 47

Same stock targets

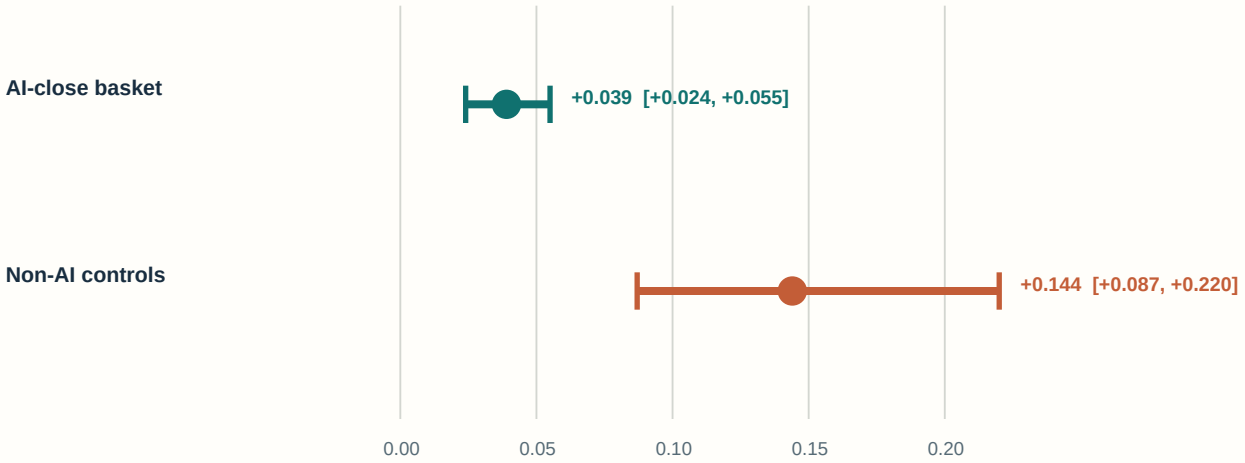
NEXT: AUTOMATE FACTOR SELECTION

Search candidate sets from 5 to 12 factors, then compare fit and selection stability across stocks and time windows. Keep selection and orthogonalization inside training data; evaluate on untouched holdouts.

Why did AI-close factors add more R-squared for controls?

When the AI-close block is added (SEMI, SOFT, ROBO, MOM), its measured gain is larger for non-AI controls.

INCREMENTAL HOLDOUT R-SQUARED (M3 - M2)



AI-close basket: NVDA, AVGO, MSFT, AMZN, GOOGL, META, ORCL

NON-AI CONTROL STOCKS

KO PG WMT MCD
JNJ JPM XOM CAT

A clue, not a conclusion

The gain may reflect general momentum or style exposure rather than AI-specific movement. Direct ETF holdings may also mechanically raise fit.

Next: isolate MOM, remove ETF overlap, and repeat across periods.

95% block-bootstrap intervals are shown for each gain; they are not a paired confidence test of the difference between groups. R-squared measures fit, not causation.

Thank you for your attention

Questions and discussion